
Genotype (deleted gene)
CGT pooled backbone context h
$E_g = \gamma_g E_g^0 e^{r \tanh(w^\top h_g)}$
$\gamma_g = 0$ for the deleted gene
κ_{cat} table
Yeast9 GEM
S, $[\ell_j, u_j]$, GPR
$\text{cap}_j = \sum_{u \in \text{GPR}(j)} \kappa_{cat,u} E_u$
$v = \Pi_{Sv=0}(\text{clip}(v_z; \ell_j, u_j, \pm \text{cap}_j))$
K alternations
$v_{\text{growth}} = v_{r_2111}$
$\lambda \sum_j \text{relu}(v_j \Delta_r G_j), \quad \lambda \in \{0, 0.1\}$
$\ell = a\, v_{\text{growth}} + b$
BCE(ℓ, $y_{\text{essential}}$)